

Erdős Problem #162

EP-162

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $\alpha > 0$ and $n \geq 1$. Let $F(n, \alpha)$ be the largest k such that there exists some 2-colouring of the edges of K_n in which any induced subgraph H on at least k vertices contains more than $\alpha \binom{|H|}{2}$ many edges of each colour.

Prove that for every fixed $0 \leq \alpha \leq 1/2$, as $n \rightarrow \infty$, $F(n, \alpha) \sim c_\alpha \log n$ for some constant c_α .

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>